

YINGKOU, China

WMO: 544710

Lat: 40.6656N

Lon: 122.1733E

Elev: 13

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.5	4.3	-15.1	2.4	7.5	-11.0	3.0	10.2	22.3	26.0	20.0	20.8	5.4	40	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	87.2	75.9	85.2	75.0	83.6	74.2	79.4	84.3	77.9	82.8	76.6	81.5	9.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	145.4	83.1	76.4	137.6	81.7	75.0	131.3	80.4	43.1	84.5	41.6	83.1	40.2	81.6	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.8	19.0	16.8	-2.4	90.3	5.9	2.3	-6.6	91.9	-10.1	93.3	-13.3	94.6	-17.6	96.3		
			-3.5	80.7	5.6	1.7	-7.5	82.0	-10.8	83.0	-13.9	83.9	-17.9	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	17.8	25.0	37.1	51.9	63.7	72.2	77.8	76.5	67.4	53.5	36.6	23.0	(6)
(7)		DBStd	21.93	7.92	8.01	7.47	6.77	5.26	4.12	3.34	4.05	5.49	7.80	9.67	8.30	(7)
(8)		HDD50	3460	998	699	402	56	0	0	0	0	55	413	837		(8)
(9)		HDD65	6484	1463	1119	864	394	87	3	0	38	361	853	1302		(9)
(10)		CDD50	3593	0	0	4	114	425	667	863	823	523	164	10	0	(10)
(11)		CDD65	1139	0	0	0	1	47	220	398	358	111	4	0	0	(11)
(12)		CDH74	7731	0	0	0	5	163	1038	3225	2815	481	4	0	0	(12)
(13)		CDH80	1570	0	0	0	0	16	155	737	618	44	0	0	0	(13)
(14)	Wind	WSAvg	7.9	6.9	7.7	9.2	10.0	9.1	7.7	7.0	6.8	6.9	7.8	8.1	7.1	(14)
(15)	Precipitation	PrecAvg	26.6	0.3	0.3	0.5	1.5	2.1	2.9	7.2	6.3	3.0	1.6	0.7	0.4	(15)
(16)		PrecMax	43.9	1.2	1.2	1.3	5.1	6.4	7.6	21.8	16.9	5.2	5.3	2.2	1.5	(16)
(17)		PrecMin	15.2	0.0	0.0	0.0	0.2	0.6	0.6	1.4	0.6	0.1	0.0	0.0	0.0	(17)
(18)		PrecStd	6.5	0.3	0.3	0.4	1.2	1.3	1.7	4.4	3.9	1.3	1.3	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	46.4	60.1	73.5	82.3	87.0	90.3	89.0	84.1	73.6	61.5	47.0	(19)
(20)		0.4%	MCWB	34.3	37.4	48.3	57.2	63.3	69.3	77.4	77.7	71.6	64.2	53.7	42.6	(20)
(21)		2%	DB	35.9	43.1	55.0	68.7	78.0	83.8	87.9	86.6	81.0	70.5	58.0	42.2	(21)
(22)		2%	MCWB	31.5	36.2	44.9	54.0	61.6	69.0	76.8	76.1	69.4	61.8	51.8	38.2	(22)
(23)		5%	DB	33.3	40.1	51.5	65.5	75.3	81.5	85.8	84.9	78.9	67.9	55.1	38.9	(23)
(24)		5%	MCWB	29.3	33.9	42.5	52.7	60.5	68.5	76.3	75.6	68.2	60.1	49.4	35.0	(24)
(25)		10%	DB	30.6	37.2	48.2	62.5	72.8	79.4	83.8	83.4	76.7	65.4	51.9	36.1	(25)
(26)		10%	MCWB	27.1	32.1	40.0	51.5	59.7	68.0	75.3	75.0	67.1	58.0	46.7	32.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.3	40.1	49.5	60.2	67.3	74.3	82.1	81.0	75.7	66.6	55.6	43.1	(27)
(28)		0.4%	MCDWB	38.1	44.6	57.9	67.9	76.6	81.2	86.1	85.4	80.7	71.8	60.0	45.7	(28)
(29)		2%	WB	32.0	37.1	46.3	57.0	65.0	72.4	79.9	79.5	73.0	63.8	52.6	38.5	(29)
(30)		2%	MCDWB	35.2	41.4	53.3	65.6	73.7	79.4	85.0	84.1	78.2	68.6	57.2	41.5	(30)
(31)		5%	WB	29.8	34.7	43.6	54.7	63.3	71.1	78.2	78.2	71.1	61.6	50.1	35.5	(31)
(32)		5%	MCDWB	33.0	39.2	49.9	62.9	71.7	78.1	83.4	82.9	76.3	66.6	54.5	38.5	(32)
(33)		10%	WB	27.4	32.3	41.0	52.5	61.7	69.9	76.8	77.0	69.3	59.3	46.9	32.8	(33)
(34)		10%	MCDWB	30.5	37.0	47.4	60.6	70.0	76.9	82.0	81.7	74.6	64.5	51.4	36.0	(34)
(35)	Mean Daily Temperature Range		MDBR	14.8	14.3	13.5	14.2	13.5	11.2	9.9	11.0	14.8	15.2	14.0	13.7	(35)
(36)		5% DB	MCDBR	16.4	16.0	16.2	17.8	16.6	14.3	12.2	12.2	16.0	15.4	16.0	15.5	(36)
(37)		5% DB	MCWBR	13.8	11.0	10.0	8.9	7.5	6.0	5.4	5.6	7.8	10.1	12.7	13.6	(37)
(38)		5% WB	MCDBR	15.9	14.4	14.5	15.2	13.4	11.4	9.8	9.2	12.5	13.7	15.0	15.1	(38)
(39)		5% WB	MCWBR	14.1	11.2	10.1	9.0	7.4	5.9	5.3	5.6	8.4	10.8	13.2	13.8	(39)
(40)		Clear-Sky Solar Irradiance	taub	0.391	0.466	0.514	0.614	0.649	0.741	0.729	0.637	0.566	0.573	0.479	0.411	(40)
(41)	taud		2.005	1.835	1.793	1.652	1.631	1.525	1.585	1.744	1.828	1.737	1.890	2.003	(41)	
(42)	Ebn at Noon		227	224	230	217	214	195	195	210	214	189	192	206	(42)	
(43)	Edh at Noon		39	54	62	77	80	89	83	69	60	59	44	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	747	1001	1431	1662	1890	1683	1531	1516	1400	1047	755	638	(44)	
(45)		RadStd	62	68	97	104	164	157	131	115	90	75	74	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	-1.82	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page